Indian Institute of Information Technology, Bhopal

Peer-to-Peer Offline Decentralized Mesh Network

Off-grid emergency SOS communication via Android Wi-Fi Aware and multi-protocol MANET/DTN simulation in NS-3.48

Semester V

SUBMITTED BY

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UNDER THE SUPERVISION OF

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Introduction

SIMULATED DISASTER GRID

12 Trapped Victims,
18 Civilian Relays,
10 SAR Foot Squads,
5 SAR Command Vehicles

Heterogeneous mobility across a 500m × 500m post-disaster zone in NS-3.48 [1]

Dual-platform approach combining an Android offline SOS app with multi-protocol C++ simulation.

Infrastructure Blackout: Disasters destroy 4G/5G towers and power grids.

MeshSOS Android App: Offline P2P distress messaging via Wi-Fi Aware[2] and Bluetooth. Utilizes a local Room database to maintain messages during temporary disconnections before forwarding them over TCP Packets.

Multi-Protocol MANET: Evaluates 9 routing schemes under Raspberry Pi (200m physical radio range) and ESP32 (60m physical radio range) limits.

DTN Store-Carry-Forward: Binary Spray & Wait bridges physical radio dead zones.

Zero-Conf & Watchdog: FNV-1a MAC-to-IPv4 hashing and Blackhole relay isolation.

Literature Review & Background

Paradigm	Protocols	Core Mechanism	Effects in a Disaster Zone
Layer-3 Reactive	`AODV[3], DSR`	On-demand RREQ/RREP path discovery	Broadcast storms and header bloat, DSR suffers from stale route caches.
Layer-3 Proactive	`OLSR[4], DSDV`	Periodic MPR and table updates	3–5s convergence lag and battery drain
Layer-2 MAC Mesh	`802.11s HWMP[5]`	Airtime Link Metric and PMP handshakes	Stateful peer links fail under mobility as HWMP was created specifically for static mesh networks
Geographic & Hybrid	`GRP/GPSR[6], ZRP`	Coordinate greedy forwarding and local zones	Requires fresh OTA location caches
DTN Replication	`Binary Spray & Wait[7], [9]`	Ticket-bounded Store-Carry-Forward	Bridges voids without buffer flooding, chosen because Epidemic Routing causes severe memory drain and buffer blockage
Device-to-Device	`Wi-Fi Aware[2] / BT`	Routerless NAN discovery and TCP sockets	Needs custom MeshRouter for multi-hop

Problem Statement & Objectives

⚠ Core MANET Bottlenecks

End-to-End Path Assumption: Standard protocols drop 100% of packets across radio voids.

Blackhole Interception: Compromised relays silently drop emergency SOS payloads.

Hardware Range Deficit: Low-power ESP32 nodes (60m range) suffer severe partitioning.

Control Message Storms: High mobility triggers RREQ and TC broadcast floods.

Communication breakdown: Civilians are unable to transmit SOS coordinates and first responders lose tactical coordination during an infrastructure collapse.

🎯 Key Project Objectives

UDP 9999 DTN Overlay: Binary Spray & Wait with 250-bundle FIFO queue.

Watchdog Trust Scoring: Passive overhearing (0.8s timer) isolating nodes below 0.50.

Zero-Conf IPv4 Hashing: Autonomous 10.254.X.X addressing via FNV-1a MAC hash.

MeshSOS Android App: Multi-hop forwarding with Room persistence and ACKs.

Heterogeneous Mobility Framework: Simulating 45 nodes with varying speeds and calibrating the physical layer according to realistic parameters.

Proposed Methodology

Our methodology integrates a real-world Android offline mesh client (MeshSOS) with an IEEE 802.11g physical channel simulation (12 Mbps data, 6 Mbps control). The RF model employs a Log-Distance Path Loss (n=2.5) calibrated to physical hardware bounds: a 200 m maximum range for Raspberry Pi nodes and a 60 m limit for ESP32 microcontrollers.

MeshSOS Transport & Discovery

Wi-Fi Aware[2] service discovery is used to establish direct TCP data paths, while Bluetooth Classic RFCOMM sockets serve as an alternative transport for point-to-point offline communication.

Application Reliability & Persistence

Employs a local Room SQLite database for message persistence.

45-Node Heterogeneous Mobility

Simulates realistic post-disaster movement speeds across a 500m x 500m grid, distributing nodes into 12 Trapped Victims, 18 Civilian Relays, 10 SAR Foot Squads and 5 High Speed SAR Vehicles

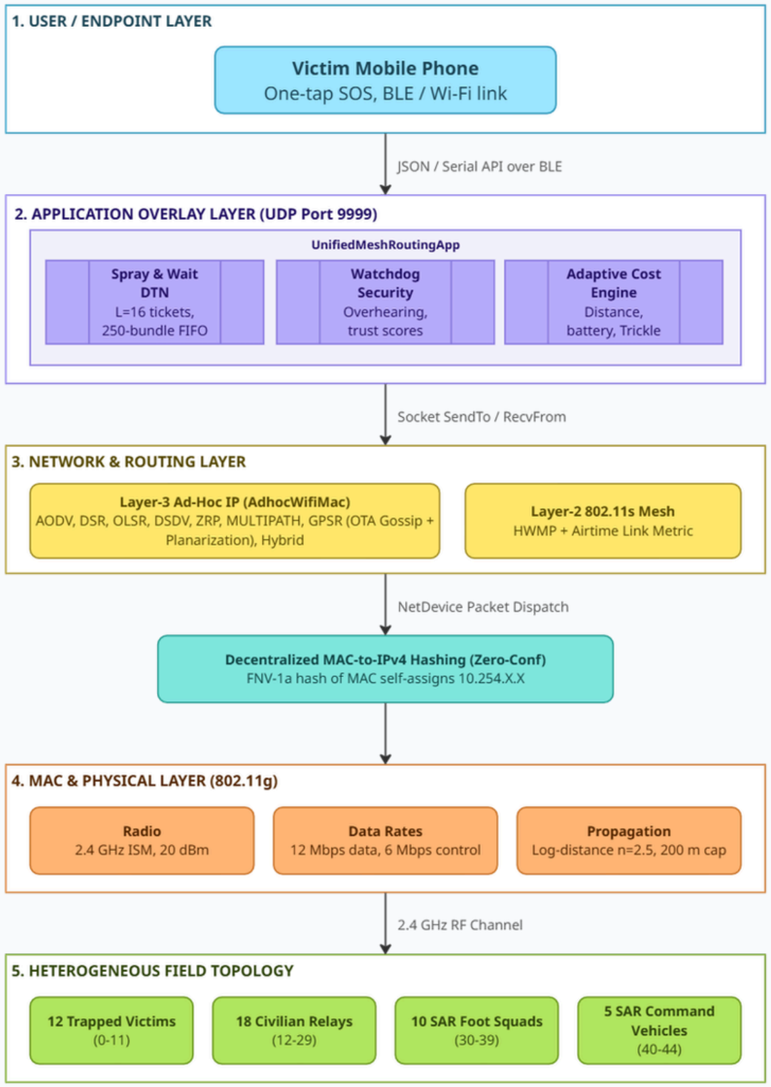
Adaptive Overhead Management

HELLO beacons dynamically adjust to local node density, scaling trickle timers from 0.5s during topology shifts up to 3.0s in stable environments to preserve node battery life.

Story-Carry-Forward DTN

Bridges physical radio voids by buffering unroutable payloads in a 250-bundle FIFO queue. Nodes logically split 16 routing tickets upon peer encounters, physically carrying messages across gaps[7], [9].

System Diagram & Algorithm



System Architecture Block Diagram

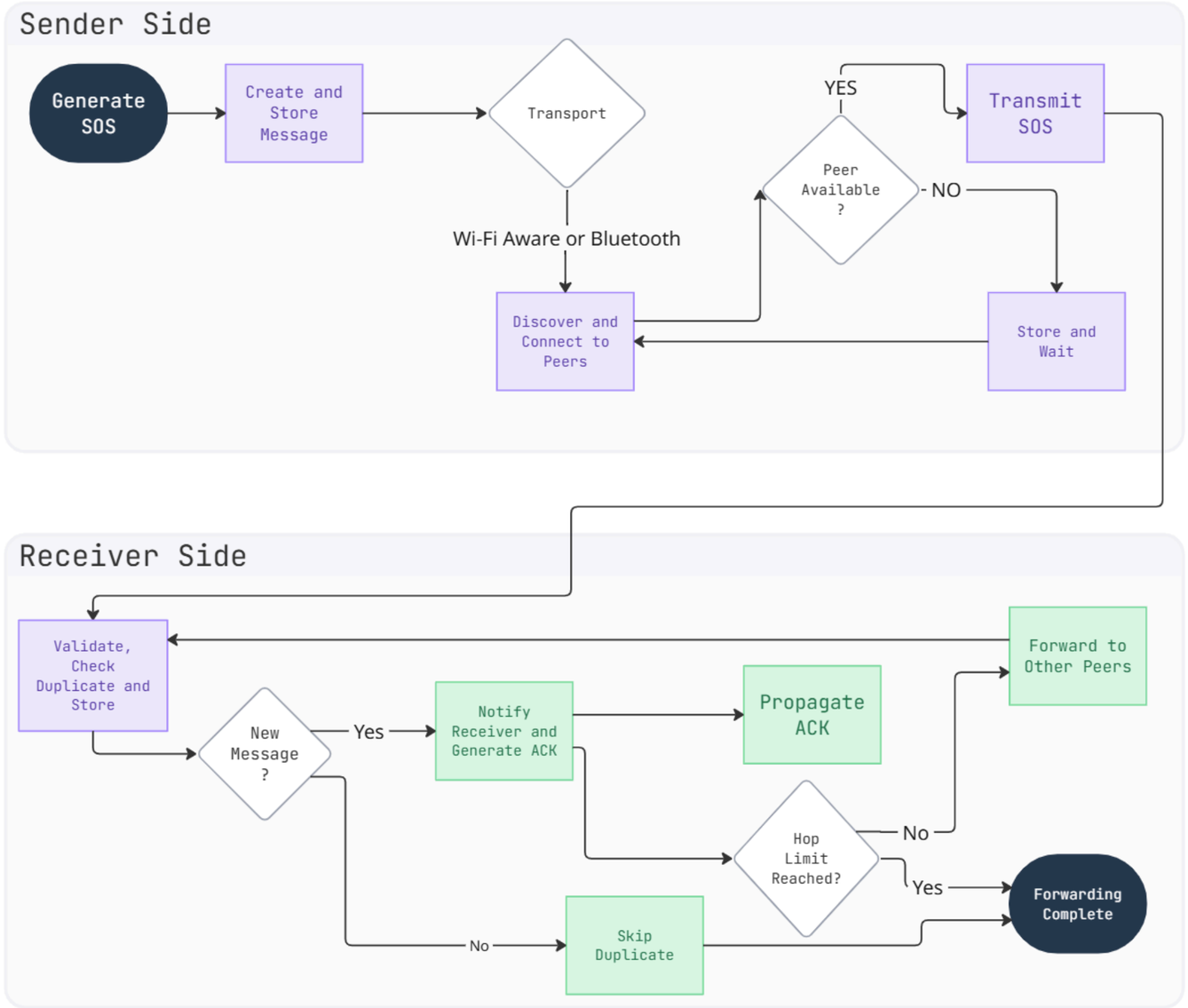
Proposed Forwarding Algorithms

Algorithm 1: Unified Queue Processing: Evaluates paths every 0.25 seconds. If direct 1-hop delivery, active MANET routes, or geographic planarization fail, it triggers the DTN fallback

Algorithm 2: Binary Spray & Wait DTN: Initializes payloads with L=16 tickets. In the "Spray" phase, a node halves its tickets with encountered peers. When L=1, it enters the "Wait" phase, physically carrying the bundle across radio dead zones [7], [9].

Algorithm 3: Offline Android Forwarding: Manages MeshSOS client logic by intercepting incoming SOS payloads, utilizing a deduplication mutex against the local database, updating hop limits, and propagating Delivery ACKs.

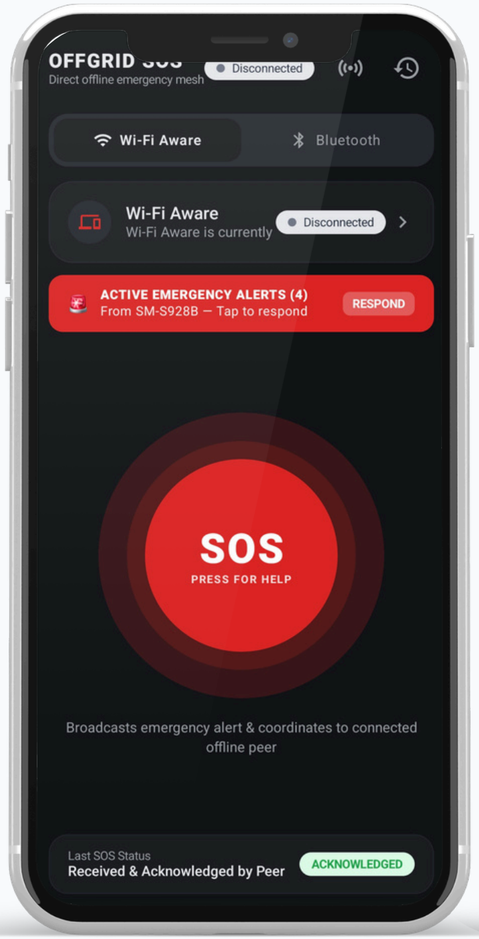
Implementation

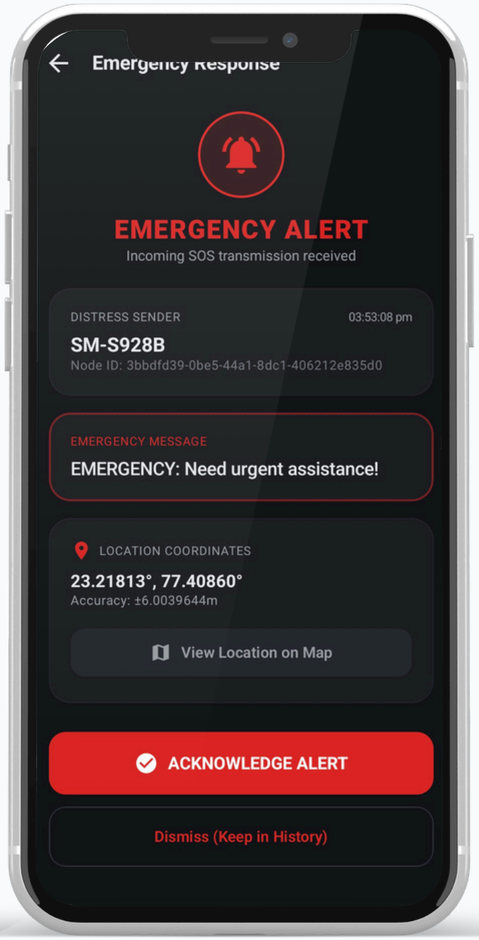


Our Works

GitHub
You can see the Source Code of MeshSOS here
Click here

Proof Documents
Consolidated proof documents of all the work
Click here





Results, Comparison & Analysis

RASPBERRY PI
(200 m radio range)

Recommended configuration

OLSR + Full Stack

Packet Delivery Ratio (PDR)	Aggregate application throughput
99.50%	**57.78 Kbps**

Near-perfect packet delivery with high application throughput, making OLSR the strongest overall Full Stack configuration on Raspberry Pi.

ESP32 MICROCONTROLLER
(60 m radio range)

Recommended configuration

HYBRID + Full Stack

Packet Delivery Ratio (PDR)	Aggregate application throughput
62.75%	**37.77 Kbps**

HYBRID delivers the strongest combined Full Stack PDR and throughput among the tested ESP32 configurations, despite constrained radio range and hardware resources.

Full Stack Performance of Recommended Configurations



Qualification: On ESP32, HYBRID +DTN alone achieves a higher PDR of 67.75% than the Full Stack configuration (62.75%). Therefore, Full Stack offers the recommended combined-performance configuration, but it does not maximize PDR in every case.

Protocol selection must balance delivery reliability, throughput, latency, and hardware constraints.

Conclusion & Future Work

Key Empirical Conclusions

Hardware Dictates Viability

ESP32 requires DTN overlay for survival.

Anycast Spatial Diversity

SOS Anycast outperforms point-to-point SAR unicast.

Autonomous Resilience

FNV-1a Zero-Conf and Watchdog prevent drops.

Future Scope & Extensions

Q-Learning Ticket Split

RL-driven dynamic DTN ticket allocation.

ESP32 + LoRa SX1262

Long-range physical transceiver deployment.

MeshSOS Offline Maps

Real-time responder tracking and voice payloads.

References

[1] NS-3.48: ns-3 Project, “ns-3 Model Library, Release 3.48,” 2026.

[2] Wi-Fi Aware: Android Developers, “Wi-Fi Aware Overview,” 2026.

[3] AODV: C. E. Perkins & E. M. Royer, IEEE WMCSA, 1999.

[4] OLSR: T. Clausen & P. Jacquet, IETF RFC 3626, 2003.

[5] IEEE 802.11s: G. R. Hiertz et al., IEEE Wireless Comm., 2010.

[6] GPSR: B. Karp & H. T. Kung, ACM MobiCom, 2000.

[7] Spray & Wait: T. Spyropoulos et al., ACM SIGCOMM WDTN, 2005.

[8] Watchdog: S. Marti et al., ACM MobiCom, 2000.

[9] DTN Architecture: K. R. Fall, ACM SIGCOMM, 2003.

[10] Hardware: Raspberry Pi Ltd & Espressif ESP32 Datasheets, 2026.

Thank You

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